Advanced Data Science · Project Pre-Submission

1. PROJECT TITLE

Decoding Citi Bike Demand: The Impact of Weather, Seasonality and Holidays in NYC

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Code: github.com/harshitkumar24cse/citibike-demand

2. STAKEHOLDER AND PROBLEM

Stakeholder: Citi Bike operations and rebalancing team (NYC). **Problem:** demand swings with weather, season and holidays, but bikes are rebalanced on fixed routines, so stations run empty or full. **Why it matters:** this causes lost trips and wasted truck runs; knowing how demand shifts lets the team move bikes ahead of time.

3. RESEARCH QUESTIONS AND HYPOTHESES

RQ1 How much do weather and the calendar change hourly demand?

RQ2 Do members and casual riders respond differently?

- **H1** Warmer hours have more trips.
- **H2** Rainy hours have fewer trips than dry hours at similar temperatures.
- **H3** Casual riders are more weather-sensitive than members.
- **H4** Casual trips are longer than member trips.
- **H5** Weekends have weaker commute peaks and a higher casual share than weekdays.

4. DATA SOURCES

Source and link	Coverage and rows	Key variables
Citi Bike trips · citibikenyc.com/system-data	Apr, Jun, Oct, Dec 2025; 15.3 M trips, 25% sample	Times, stations, coordinates, bike type, rider type
Open-Meteo weather · open-meteo.com	Same months, 2,928 hours	Temperature, rain, wind, humidity, pressure, cloud
NOAA daily records · ncei.noaa.gov	Same months, 122 days	Max/min temperature, rain (validation only)
Citi Bike GBFS stations · gbfs.citibikenyc.com	~2,500 stations (current)	Capacity, location
US/NY holidays · Python holidays	14 in 2025 (3 in study months)	Holiday date and name

Rows: 3,822,094 trips × 41 variables. **Constraints:** ~3 GB of trip files (seeded 25% sample for 8 GB RAM); no NOAA hourly data for Dec 2025, so Open-Meteo is used throughout and checked against NOAA (temperature r = 0.99, wet days agree 92%); station data is a current snapshot (96.6% match).

5. INTEGRATION PLAN (SSOT)

- **Keys:** one row per trip; start hour → weather; station ID → capacity; date → holiday.
- **Cleaning:** duplicate ride IDs dropped (0); trips under 1 or over 180 min (4,410) and outside the study months (361) removed.
- **Missing values:** weather gaps up to 3 h interpolated; unmatched stations and coordinates left empty.
- **Units:** °C, mm, m/s, hPa, %, km, min, km/h. **Checks:** row count unchanged after joins; unmatched weather 0%, stations 3.4%.

6. METHODS AND BADIR WORKFLOW

Business question	When, and by how much, do weather and the calendar shift demand?
Analysis plan	Trips per hour; EDA and correlation; t-test, Mann-Whitney U, Kruskal-Wallis, chi-square; multiple regression on weather and calendar.
Data collection	build_ssot.py pipeline builds the SSOT (done).
Insights	Effect sizes with confidence intervals; hypotheses rated supported, not supported or inconclusive.
Recommendations	Rebalancing and staffing rules for the operations team.

7. VALIDATION AND DECISION VALUE

Each hypothesis gets a test with effect size, a chart and a robustness check (per month, weekdays only, non-parametric).
Decision value: change in trips per °C, rainy hour and day type → extra rebalancing before warm, dry weekends; lighter dispatch in rain or freezing cold; priority for large stations.

8. LIMITATIONS AND NEXT STEPS

- **Assumptions:** one weather point for the city; under 0.1 mm/h counts as dry; 1-180 min trips are genuine.
- **Constraints:** one month per season; trips understate demand at empty or full stations; station snapshot; only 3 holidays (described, not tested); association, not causation.
- **Milestones:** Stage 1 problem, data, SSOT (done) → Stage 2 quality report, EDA, tests → Stage 3 findings, recommendations, roadmap.